INF6422E Advanced Concepts in Computer Security

Practical Work 4 – Winter 2026

Semantic and Context-Aware Security Analysis

Instructions:

- This work is to be submitted in a group of two or three and via Moodle only.
- The submitted report must be in pdf form and also include (.py, .ipynb) with it. You're free to build it in any format you want, however (.docx, .odt, .tex, etc.).
- The report must contain a title page including the course title, the lab title, your names and student ID numbers (Matricule).
- The report must be submitted by the 31st of March 2026 before 23h59. A penalty of 10% will be applied for each day after that date.
- There are 4 labs in total, each of 20 marks/10 weightage.

Objective:

This lab focuses on semantic, contextual, and trustworthy interpretation of cybersecurity alerts using a modern intrusion detection dataset (UNSW-NB15) and a gradient boosting model (XGBoost). This lab emphasizes advanced multi-class attack analysis, semantic interpretation of alerts, and context-aware security reasoning rather than only improving prediction accuracy. Students will analyze how contextual network features (protocol, service, load, timing, and attack category) influence intrusion behavior and how machine learning outputs can be interpreted in realistic Security Operations Center (SOC) environments.

Key Learning Goals:

1. Understand multi-class intrusion detection using modern cybersecurity datasets.
2. Apply XGBoost for complex attack category classification.

3. Perform context-aware and semantic interpretation of security alerts.
4. Analyze misclassifications from a cybersecurity perspective (not only metrics).
5. Connect machine learning outputs to trustworthy decision-making in SOC environments.

Dataset:

UNSW-NB15 Intrusion Detection Dataset

Link: https://research.unsw.edu.au/projects/unsw-nb15-dataset

Or

https://drive.google.com/drive/folders/19fuKj2Ij9Z-kgmTo61TWyLX1JVyYrveO?usp=sharing

The dataset contains modern network traffic with rich contextual features, including:

- Protocol (proto)
- Service type (service)
- State information (state)
- Temporal features (Stime, Ltime)
- Traffic behavior (packets, bytes, load, jitter)
- Attack categories (attack_cat)
- Binary label (Label: 0 = normal, 1 = attack)

1. Comparative Security Modeling [6 points]

1.1 Binary Intrusion Detection

Train an XGBoost classifier to predict:

- Label (0 = Normal, 1 = Attack)

You must:

- Handle categorical features (proto, service, state)
- Justify preprocessing decisions

Deliverable:

- Description of preprocessing and encoding strategy
- Confusion Matrix
- Precision, Recall, F1-score
- False Positive Rate (FPR) and False Negative Rate (FNR)

1.2 Class Attack Category Classification

Using the same dataset, train a second XGBoost model to predict:

- attack_cat (e.g., DoS, Exploits, Reconnaissance, etc.)

This task reflects real-world IDS where identifying the type of attack is more difficult than simple detection.

Deliverable:

- Multi-class confusion matrix
- Per-class recall and F1-score
- Short discussion:
    - Why multi-class attack detection is harder than binary detection
    - Which attack categories are most difficult to classify and why

2. Context-Aware Feature Semantics [6 points]

2.1 Group the dataset features into meaningful cybersecurity categories such as:

- Network Context (proto, service, state)
- Traffic Behavior (Sload, Dload, Spkts, Dpkts)
- Temporal Context (dur, Stime, Ltime)
- Connection Patterns (ct_* features)

Deliverable:

A structured table:

Feature Group	Example Features	Security Meaning	Role in Attack Detection

- Plus a short explanation of how contextual features help distinguish attacks from normal traffic.

2.2 Attack Behavior Profiling

Select at least three attack categories (e.g., DoS, Exploits, Reconnaissance) and analyze their behavior using:

- Load features
- Packet features
- Duration
- Service usage

Deliverable:

- Comparative analysis of selected attack categories
- Plots or statistics
- Security interpretation explaining how each attack differs from benign traffic

3. Misclassification and Security Error Analysis [5 points]

3.1 Misclassification and Security Error Analysis

Identify attack samples classified as normal and analyze their contextual characteristics.

Deliverable:

- Examples of false negative samples
- Short explanation:
    - Do stealth attacks mimic benign traffic?
    - Why does the model fail to detect them?

3.2 False Positive Security Interpretation

Analyze benign samples classified as attacks and interpret them from an operational IDS perspective.

Deliverable:

- Examples of false positive samples
- Short discussion:
    - Do high-load benign flows resemble attacks?
    - How do false alarms affect SOC analysts?

4. Trustworthy Security Interpretation (SOC Perspective) [3 points]

4.1 Security-Centric Evaluation (Beyond Accuracy)

Critically discuss why high accuracy alone is not sufficient for cybersecurity systems.

You must consider:

- Impact of False Negatives (missed attacks)
- Impact of False Positives (alert fatigue)
- Trust in automated ML-based IDS alerts

Deliverable:

- A short (1 page maximum) reflection on:
    - Reliability of ML in cybersecurity
    - Practical deployment challenges in SOC environments
    - Limitations of model-based intrusion detection

Resources:

https://scikit-learn.org/stable/

https://xgboost.readthedocs.io/en/release_3.2.0/

https://github.com/dmlc/xgboost

https://gist.github.com/pb111/cc341409081dffa5e9eaf60d79562a03

References:

1. A. Husain, A. Salem, C. Jim and G. Dimitoglou, "Development of an Efficient Network Intrusion Detection Model Using Extreme Gradient Boosting (XGBoost) on the UNSW-NB15 Dataset," 2019 IEEE International Symposium on Signal Processing and Information Technology (ISSPIT), Ajman, United Arab Emirates, 2019, pp. 1-7, doi: 10.1109/ISSPIT47144.2019.9001867.

2. N. Pansari, S. Srivastava, R. R H and M. Agarwal, "Attack Classification using Machine Learning on UNSW-NB 15 dataset using XGBoost Feature Selection & Ablation Analysis," 2024 IEEE 9th International Conference for Convergence in Technology (I2CT), Pune, India, 2024, pp. 1-9, doi: 10.1109/I2CT61223.2024.10543523.

3. U. C. Akuthota and L. Bhargava, "Network Intrusion Classification on the UNSW-NB15 Dataset Using XGBoost Feature Selection Technique," 2023 IEEE International Symposium on Smart Electronic Systems (iSES), Ahmedabad, India, 2023, pp. 169-174, doi: 10.1109/iSES58672.2023.00043.

4. P. Ferreira, E. Martins, J. Silva and P. Teixeira, "Feature Selection and XGBoost for Enhanced Intrusion Detection: A Comparative Study Across Benchmark Datasets," 2025 13th International Symposium on Digital Forensics and Security (ISDFS), Boston, MA, USA, 2025, pp. 1-6, doi: 10.1109/ISDFS65363.2025.11012060.